

File Edit Selection View Go Run Terminal Help

MLOPS

EXPLORER

1st code experiment.ipynb 2nd_mlfow_binaryclassification.ipynb X 3rd_mlfow_model_registration.ipynb meta.yaml

mirruns > trash > 2nd_mlfow_binaryclassification.ipynb > Experiment 4: Handle class imbalance using SMOTETomek and then Train XGBoost > empty cell

Generate + Code + Markdown Run All Restart Clear All Outputs Jupyter Variables Outline

Python

OPEN EDITORS

1st code experiment.ip... 2nd_mlfow_binaryclas... 3rd_mlfow_model_re... meta.yaml mirruns0

MLOPS

.venv mlartifacts mirruns trash mirruns 1st code experiment.ipynb 2nd_mlfow_binaryclassifi... 3rd_mlfow_model_regis... 0 meta.yaml 46428833761025439 743607908228897555 904159297191855621 models

import mlflow.xgboost

[12] ✓ 12s

Initialize MLflow
mlflow.set_experiment("imbalanced classification")
mlflow.set_tracking_uri("http://localhost:5000")

for i, element in enumerate(models):
 model_name = element[0]
 model = element[1]
 report = reports[i]

 with mlflow.start_run(run_name=model_name):
 mlflow.log_param("model", model_name)
 mlflow.log_metric('accuracy', report['accuracy'])
 mlflow.log_metric('recall_class_1', report['1']['recall'])
 mlflow.log_metric('recall_class_0', report['0']['recall'])
 mlflow.log_metric('f1_score_macro', report['macro avg']['f1-score'])

 if "XGB" in model_name:
 mlflow.xgboost.log_model(model, "model")
 else:
 mlflow.sklearn.log_model(model, "model")

[14]

warnings.warn(DEPRECATION_MESSAGE, DeprecationWarning)

INFO: Unicorn running on http://127.0.0.1:5000 (Press CTRL-C to quit)

INFO: Started parent process [19192]

INFO: Started server process [21656]

INFO: Waiting for application startup.

INFO: Application startup complete.

INFO: Started server process [5036]

INFO: Waiting for application startup.

INFO: Application startup complete.

INFO: Started server process [22088]

INFO: Waiting for application startup.

INFO: Application startup complete.

OUTLINE

TIMELINE

power... powershell mlflow powershell mlflow

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EXPLORER

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1st code experiment.ipynb 2nd_milflow_binaryclassification.ipynb 3rd_milflow_model_registration.ipynb meta.yaml

mlruns > trash > 3rd_milflow_model_registration.ipynb > import numpy as np

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1st code experiment.ipynb 2nd_milflow_binaryclassification.ipynb 3rd_milflow_model_registration.ipynb meta.yaml mlruns0

MLOPS

.venv mlartifacts mlruns trash mlruns 1st code experiment.ipynb 2nd_milflow_binaryclassification.ipynb 3rd_milflow_model_registration.ipynb 0 meta.yaml 464428833761025439 743607908228897555 904159297191855621 models

2025/10/06 20:44:00 WARNING mlflow.models.model: `artifact\_path` is deprecated. Please use `name` instead.

2025/10/06 20:44:05 WARNING mlflow.models.model: Model logged without a signature and input example. Please set `input\_example` parameter.

2025/10/06 20:44:05 WARNING mlflow.models.model: `artifact\_path` is deprecated. Please use `name` instead.

🚀 View run Logistic Regression at: <http://localhost:5000/#/experiments/464428833761025439/runs/abfaa2d3c7d24116ab7166e97d5636e2>

📌 View experiment at: <http://localhost:5000/#/experiments/464428833761025439>

2025/10/06 20:44:08 WARNING mlflow.models.model: Model logged without a signature and input example. Please set `input\_example` parameter.

2025/10/06 20:44:09 WARNING mlflow.models.model: `artifact\_path` is deprecated. Please use `name` instead.

🚀 View run Random Forest at: <http://localhost:5000/#/experiments/464428833761025439/runs/c0df08d1a5e04b828a5ade6cbcf88e29>

📌 View experiment at: <http://localhost:5000/#/experiments/464428833761025439>

2025/10/06 20:44:14 WARNING mlflow.models.model: Model logged without a signature and input example. Please set `input\_example` parameter.

2025/10/06 20:44:14 WARNING mlflow.models.model: `artifact\_path` is deprecated. Please use `name` instead.

🚀 View run XGBClassifier at: <http://localhost:5000/#/experiments/464428833761025439/runs/dc0ace693941408ea8b940f2acdd0e4b>

📌 View experiment at: <http://localhost:5000/#/experiments/464428833761025439>

2025/10/06 20:44:18 WARNING mlflow.models.model: Model logged without a signature and input example. Please set `input\_example` parameter.

🚀 View run XGBClassifier With SMOTE at: <http://localhost:5000/#/experiments/464428833761025439/runs/511345bf4a6a4678b53ec0919c33647>

📌 View experiment at: <http://localhost:5000/#/experiments/464428833761025439>

Register the Model

```
[14]: model_name = 'XGB-Smote'
```

warnings.warn(DEPRECATION_MESSAGE, DeprecationWarning)

INFO: Unicorn running on http://127.0.0.1:5000 (Press CTRL-C to quit)

INFO: Started parent process [19192]

INFO: Started server process [21656]

INFO: Waiting for application startup.

INFO: Application startup complete.

INFO: Started server process [5036]

INFO: Waiting for application startup.

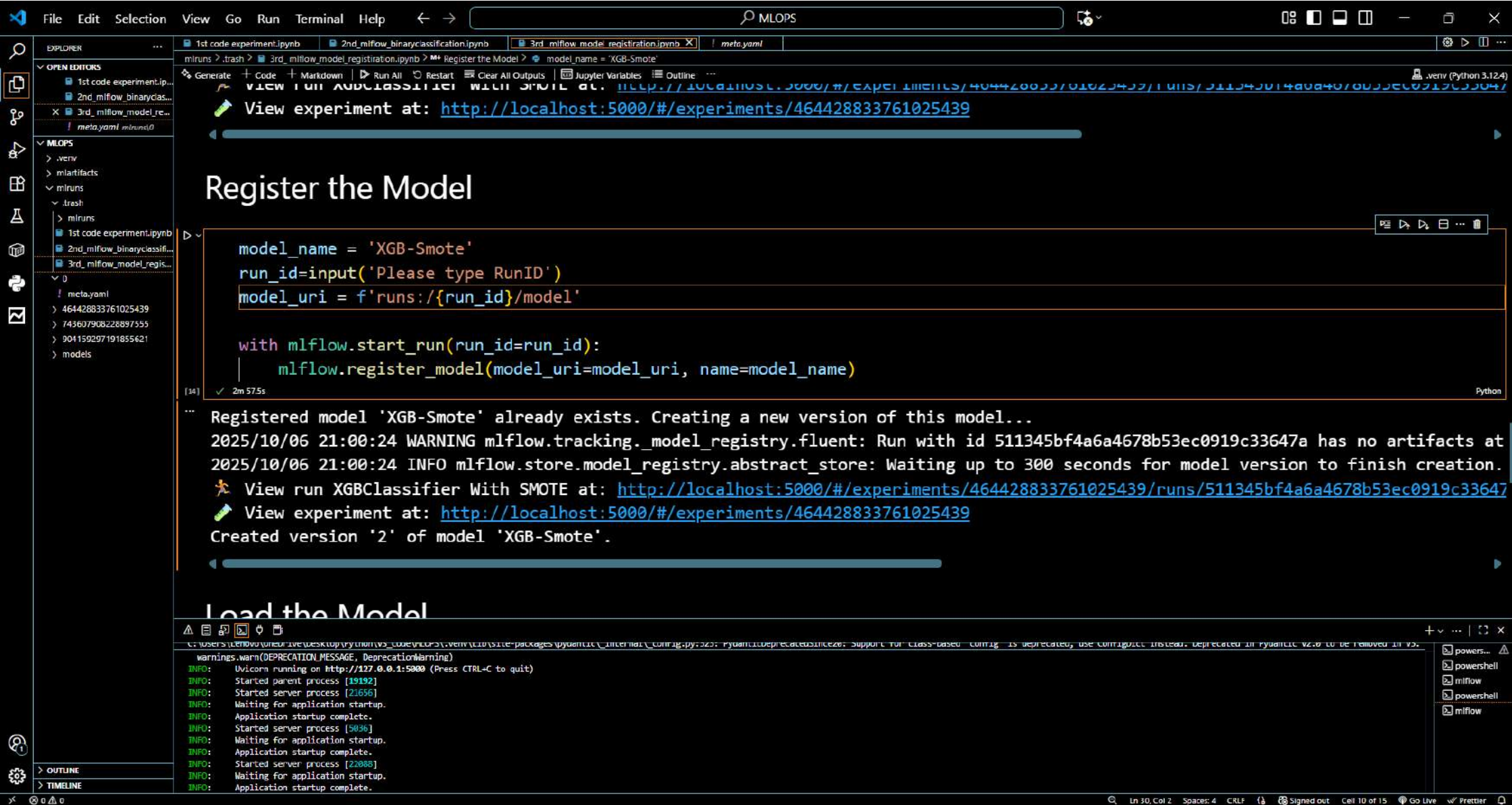
INFO: Application startup complete.

INFO: Started server process [22088]

INFO: Waiting for application startup.

INFO: Application startup complete.

Ln 30, Col 2 Spaces: 4 Spaces: 4 CRLF Signed out Cell 1 of 15 Go Live



FileEditSelectionViewGoRunTerminalHelp

MLOPS

1st code experiment.ipynb2nd_mlflow_binaryclassification.ipynb3rd_mlflow_model_registration.ipynbmeta.yaml

mlruns>.trash>3rd_mlflow_model_registration.ipynb>Register the Model>model_name = 'XGB-Smote'

GenerateCodeMarkdownRun AllRestartClear All OutputsJupyter VariablesOutline

VIEW FULL ADDRESS WITH SHORT URL: <http://localhost:5000/#/experiments/464428833761025439>

View experiment at: <http://localhost:5000/#/experiments/464428833761025439>

EXPLORE

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1st code experiment.ipynb2nd_mlflow_binaryclassification.ipynb3rd_mlflow_model_registration.ipynbmeta.yaml mlruns0

MLOPS

.venvmlartifactsmlruns.trashmlruns

1st code experiment.ipynb2nd_mlflow_binaryclassification.ipynb3rd_mlflow_model_registration.ipynb0

! meta.yaml

464428833761025439743607908228897555904159297191855621models

OUTLINE

TIMELINE

Register the Model

```
model_name = 'XGB-Smote'
run_id=input('Please type RunID')
model_uri = f'runs://{run_id}/model'

with mlflow.start_run(run_id=run_id):
```

PS C:\Users\Lenovo\OneDrive\Desktop\Python\VS_Code\MLOPS> & C:/Users/Lenovo/OneDrive/Desktop/Python/VS_Code/MLOPS/.venv/Scripts/Activate.ps1
(.venv) PS C:\Users\Lenovo\OneDrive\Desktop\Python\VS_Code\MLOPS> mlflow ui
C:\Users\Lenovo\OneDrive\Desktop\Python\VS_Code\MLOPS\.venv\Lib\site-packages\pydantic_internal_config.py:323: PydanticDeprecatedSince20: Support for class-based `config` is deprecated, use ConfigDict instead. Deprecated in Pydantic V2.0 to be removed in V3.
0. See Pydantic V2 Migration Guide at https://errors.pydantic.dev/2.11/migration/
warnings.warn(DEPRECATION_MESSAGE, DeprecationWarning)
INFO: Uvicorn running on http://127.0.0.1:5000 (Press CTRL+C to quit)
INFO: Started parent process [19192]
INFO: Started server process [21656]
INFO: Waiting for application startup.
INFO: Application startup complete.
INFO: Started server process [5036]
INFO: Waiting for application startup.
INFO: Application startup complete.
INFO: Started server process [22088]
INFO: Waiting for application startup.
INFO: Application startup complete.
INFO: Started server process [8404]
INFO: Waiting for application startup.
INFO: Application startup complete.

powershellmlflowpowershellmlflow

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MLflow

MLflow

MLflow

MLflow

localhost:5000/#/experiments

mlflow 3.4.0

GitHub

Docs

+ New

Experiments

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Experiments

Create

Compare

Delete

Filter experiments by name

Q

Tag filter

☐	Name	Time created	Last modified	Description	Tags
☐	deployment model	10/06/2025, 08:44:00 PM	10/06/2025, 08:44:00 PM	-	
☐	imbalanced classification	10/06/2025, 07:48:19 PM	10/06/2025, 07:48:19 PM	-	
☐	1st Experiment	10/06/2025, 07:32:45 PM	10/06/2025, 07:32:45 PM	-	
☐	Default	10/01/2025, 08:17:38 PM	10/01/2025, 08:17:38 PM	-	

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MLflow

MLflow

MLflow

MLflow

localhost:5000/#/models

mlflow 3.4.0

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Experiments

Models

Prompts

Registered Models

Share and manage machine learning models. [Learn more](#)

Filter registered models by name or ...

Create Model

Name	Latest version	Aliased versions	Created by	Last modified	Tags
XGB-Smote	Version 2	@ appserver : Version 1		10/06/2025, 09:00:24 ...	—
finalproduction	Version 1			10/06/2025, 08:55:28 ...	—

New model registry UI

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